

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

张杰博士生于 1990 年 10 月 20 日，现任北京大学计算机学院助理教授、特聘研究员、博士生导师，获得优秀青年科学基金海外项目（海外优青）、北京市自然科学基金青年重点项目（原北京杰青）、华为奥林帕斯先锋奖、英特尔学术英才计划荣誉学者、ACM SIGCSE 新星奖、博雅青年学者。他长期从事存储系统和专用处理器的研究和设计，致力于从计算机体系结构层面出发，解决大数据和人工智能时代对于高性能存储系统的需求，突破冯诺依曼体系结构下数据迁移的瓶颈以及内存墙的限制。负责和参与科技部重点研发项目（课题负责人）、国家自然科学基金重点项目（北大负责人）、国家科技重大专项和面上项目，得到华为、江波龙、三星电子、海力士、英特尔和西部数据的资助。自 2016 年起，在计算机国际会议及期刊上发表了 60 余篇论文，包括 ISCA、OSDI、SOSP、HPCA、MICRO、ASPLOS、FAST、ATC、NSDI、DAC 和 Eurosys。2025 年入选 HPCA 名人堂（首位 90 后成员）。在计算机系统结构领域 90 后学者中，发表顶级会议论文数排名全球第一（CSRankings, 2016-2026）。亚洲历史上第三位在所有体系结构和系统顶级会议上发表论文的学者。论文获 2019 年韩国计算机大会（KCC）最佳演讲论文奖和 2019 年非易失性内存研讨会（NVMW）优秀论文提名奖。在存储领域的研究成果被列为 KAIST50 年重大突破之一，并被四十余家新闻媒体报道。

工作经历

北京大学，北京，中国	助理教授，博导
计算机学院	2021 年 7 月–至今
KAIST, Daejeon, Korea	博士后
Electrical Engineering (Computing Division)	2020 年 3 月 –2021 年 5 月

教育经历

Yonsei University, Incheon, Korea	Advisor: Dr. Myoungsoo Jung
PhD in Engineering	August 2015 – Feb 2020
University of Texas at Dallas, Richardson, Texas	Advisor: Dr. Myoungsoo Jung
PhD in Computer Engineering (transfer to Korea)	August 2014 – August 2015
University of Texas at Dallas, Richardson, Texas	Advisor: Dr. Myoungsoo Jung
Master of Science in Electrical Engineering	August 2012 – May 2014
Nanjing University of Posts and Telecommunications, Nanjing, China	
BS in Communication Engineering (computer communication)	September 2008 – July 2012

论文发表情况

2026

ICS	Latency-SLO-Aware Memory Offloading for Large Language Model Inference
CCF-B 会议	Chenxiang Ma, Hanyu Zhao, Zhisheng Ye, Zehua Yang, Tianhao Fu, Jiaxun Han, Jie Zhang , Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Yong Li, Diyu Zhou
	ACM International Conference on Supercomputing
HPDC	BCCE: Block-Centric GPU Co-Design for Real-Time Range-Top-K Query at Scale
CCF-A 会议	Chengying Huan, Ziheng Meng, Zhengyi Yang, Yongchao Liu, Jie Zhang , Qing Wang, Jing Wang, Shaonan Ma, Zhibin Wang, Mingxing Zhang, Rong Gu, Baokun Wang, Guihai Chen, Chen Tian
	ACM International Symposium on High-Performance Parallel and Distributed Computing
OSDI	Espresso: Constructing Cost-Efficient CXL JBOF via Inter-SSD Computing Resource
CCF-A 会议	Sharing

Shushu Yi, Yuda An, Li Peng, Xiurui Pan, Qiao Li, Jieming Yin, Guangyan Zhang, Wenfei Wu, Chenxi Wang, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Ke Zhou, **Jie Zhang***

USENIX Symposium on Operating Systems Design and Implementation

OSDI

Merlin: An Efficient Adaptive Cache Eviction Algorithm via Fine-Grained Characterization

CCF-A 会议

Liuja Li, Jinhao Guo, Yi Fan, Jianyu Wu, Zhenling Wang, **Jie Zhang**, Yuval Tamir, Xiaolin Wang, Yingwei Luo, Diyu Zhou

USENIX Symposium on Operating Systems Design and Implementation

OSDI

MultiLane: Eliminating Centralized Bottlenecks in User-space Network Stack

CCF-A 会议

Kang Hu, Shuqi Dong, Chuandong Li, Ran Yi, Zonghao Zhang, **Jie Zhang**, Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou

USENIX Symposium on Operating Systems Design and Implementation

ICLR

LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences

CCF-A 会议

Wenbo Wu, Qingyi Si, Xiurui Pan, Ye Wang, **Jie Zhang***

The Fourteenth International Conference on Learning Representations

DAC

WILL: Write Invalidation Skipping for QLC NAND by Leveraging Data Lifespan and Latency Awareness

CCF-A 会议

Beichen Ning, Yina Lv, Tianyu Ren, Bin Gao, Xiangyu Yao, **Jie Zhang**, Qiao Li and Chun Jason Xue

Design Automation Conference

DAC

Cachence: Fine-Grained Cache Partitioning in Both Time and Space

CCF-A 会议

Liuja Li, Yuanlong Li, Yiming Yao, Jianyu Wu, Yi Fan, Jinhao Guo, Liren Zhu, **Jie Zhang**, Xiaolin Wang, Yingwei Luo, Zhenlin Wang, Diyu Zhou

Design Automation Conference

FAST

Xerxes: Extensive Exploration of Scalable Hardware Systems with CXL-Based Simulation Framework

CCF-A 会议

Yuda An, Shushu Yi, Bo Mao, Qiao Li, Mingzhe Zhang, Diyu Zhou, Ke Zhou, Nong Xiao, Guangyu Sun, Yingwei Luo, **Jie Zhang***

The USENIX Conference on File and Storage Technologies

HPCA

AutoGNN: End-to-End Hardware-Driven Graph Preprocessing for Enhanced GNN

CCF-A 会议

Performance

Seungkwan Kang, Seungjun Lee, Donghyun Gouk, Miryeong Kwon, Hyunkyu Choi, Junhyeok Jang, Sangwon Lee, Huiwon Choi, **Jie Zhang**, Wonil Choi, Mahmut T Kandemir, Myoungsoo Jung

IEEE International Symposium on High-Performance Computer Architecture

Eurosys

ColdCode: Cold Data Encoding for Enhanced Reliability and Lifetime in 3D NAND Flash

CCF-A 会议

Qiao Li, Shangyu Wu, Zheng Wan, Yufei Cui, **Jie Zhang***, Chun Jason Xue

21th ACM European Conference on Computer Systems

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

TCAD	CXL-DMSim: A Full-System CXL Disaggregated Memory Simulator With Comprehensive Silicon Validation
CCF-A 期刊	Yanjing Wang, Lizhou Wu, Wentao Hong, Yang Ou, Zicong Wang, Sunfeng Gao, Jie Zhang , Sheng Ma, Dezun Dong, Xingyun Qi, Mingche Lai, Nong Xiao ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems
PACT	Exploring Memory Tiering Systems in the CXL Era via FPGA-based Emulation and Device-Side Management
CCF-B 会议	Yiqi Chen, Xiping Dong, Zhe Zhou, Zhao Wang, Jie Zhang , Guangyu Sun, International Conference on Parallel Architecture and Compilation Techniques
NEWTON	Ultrafast and Reliable Domain-Wall and Skyrmion Logic in a Chirally Coupled Ferrimagnet
CELL 子刊	Yifei Ma, Dihua Wu, Fengbo Yan, Xiaoxiao Fang, Peixin Qin, Leran Wang, Li Liu, Laichuan Shen, Zhiqi Liu, Wenyun Yang, Jie Zhang , Yanglong Hou, Yan Zhou, Feng Luo, Jinbo Yang, Kaiming Cai, Shuai Ning, Zhaochu Luo Cell Press NEWTON
SOSP	Aeolia: Fast and Secure Userspace Interrupt-Based Storage Stack
CCF-A 会议	Chuangdong Li, Ran Yi, Zonghao Zhang, Jing Liu, Changwoo Min, Jie Zhang , Yingwei Luo, Xiaolin Wang, Zhenling Wang, Diyu Zhou ACM Symposium on Operating Systems Principles
HotStorage	SPDK+: Low Latency or High Power Efficiency? We Take Both
	Endian Li, Shushu Yi, Li Peng, Qiao Li, Diyu Zhou, Zhenlin Wang, Xiaolin Wang, Bo Mao, Yingwei Luo, Ke Zhou, Jie Zhang* The 17th ACM Workshop on Hot Topics in Storage and File Systems
ICDE	BL-Tree: The Best of Both Worlds by combining B+-Tree on Top and LSM-Tree on Bottom
CCF-A 会议	Maoxin Ye, Zuocheng Wang, Shengzhe Wang, Jiahong Chen, Chunfeng Du, Jie Zhang , Ke Zhou, Suzhen Wu, Bo Mao IEEE International Conference on Data Engineering
ToS	Enhancing the Performance of Next-Generation SSD Arrays: A Holistic Approach
CCF-A 期刊	Jie Zhang* , Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung ACM Transactions on Storage
TCAD	FlexHMB: A Flexible HMB Design towards Bufferless Mobile Flash
CCF-A 期刊	Yong Peng, Li Peng, Shaocong Sun, Lujia Yin, Qiao Li, Jie Zhang* ACM Transactions on Computer-Aided Design of Integrated Circuits and Systems
ISCA	XHarvest: Rethinking High-Performance and Cost-Efficient SSD Architecture with CXL-Driven Harvesting
CCF-A 会议	Li Peng, Wenbo Wu, Shushu Yi, Xianzhang Chen, Chenxi Wang, Shengwen Liang, Zhe Wang, Nong Xiao, Qiao Li, Mingzhe Zhang, Jie Zhang* IEEE/ACM International Symposium on Computer Architecture

ISCA	ArtMem: Adaptive Migration in Reinforcement Learning-Enabled Tiered Memory
CCF-A 会议	Xinyue Yi, Hongchao Du, Yu Wang, Jie Zhang , Qiao Li, Chun Jason Xue IEEE/ACM International Symposium on Computer Architecture
Eurosys	Daredevil: Rescue Your Flash Storage from Inflexible Kernel Storage Stack
CCF-A 会议	Junzhe Li, Ran Shu, Jiayi Lin, Qingyu Zhang, Ziyue Yang, Jie Zhang , Yongqiang Xiong, Chenxiong Qian 20th ACM European Conference on Computer Systems
NSDI	Beehive: A Scalable Disaggregated Memory Runtime Exploiting Asynchrony of
CCF-A 会议	Multithreaded Programs Quanxi Li, Hong Huang, Ying Liu, Yanwen Xia, Jie Zhang , Mosong Zhou, Xiaobing Feng, Huimin Cui, Quan Chen, Yizhou Shan, Chenxi Wang 22nd USENIX Symposium on Networked Systems Design and Implementation
HPCA	InstAttention: In-Storage Attention Offloading for Cost-Effective Long-Context LLM
CCF-A 会议	Inference Xiurui Pan, Endian Li, Qiao Li, Shengwen Liang, Yizhou Shan, Ke Zhou, Yingwei Luo, Xiaolin Wang, Jie Zhang * IEEE International Symposium on High-Performance Computer Architecture
HPCA	NeuVSA: A Unified and Efficient Accelerator for Neural Vector Search
CCF-A 会议	Ziming Yuan, Lei Dai, Wen Li, Jie Zhang , Shengwen Liang, Ying Wang, Cheng Liu, Huawei Li, Jiafeng Guo, Peng Wang, Renhai Chen, Gong Zhang IEEE International Symposium on High-Performance Computer Architecture
HPCA	Criticality-Aware Instruction-Centric Bandwidth Partitioning for Data Center Applications
CCF-A 会议	Liren Zhu, Liujia Li, Jianyu Wu, Yiming Yao, Zhan Shi, Jie Zhang , Zhenlin Wang, Xiaolin Wang, Yingwei Luo, Diyu Zhou IEEE International Symposium on High-Performance Computer Architecture
2024	
SOSP	BIZA: Design of Self-Governing Block-Interface ZNS AFA for Endurance and Performance
CCF-A 会议	Shushu Yi, Shaocong Sun, Li Peng, Yingbo Sun, Ming-Chang Yang, Zhichao Cao, Qiao Li, Myoungsoo Jung, Ke Zhou, Jie Zhang * ACM Symposium on Operating Systems Principles
MICRO	NeoMem: Hardware/Software Co-Design for CXL-Native Memory Tiering
CCF-A 会议	Zhe Zhou, Yiqi Chen, Tao Zhang, Yang Wang, Ran Shu, Shuotao Xu, Peng Cheng, Lei Qu, Yongqiang Xiong, Jie Zhang , Guangyu Sun The 57 th Annual IEEE/ACM International Symposium on Microarchitecture
MICRO	Cambricon-Ilm: A chiplet-based hybrid architecture for on-device Inference of 70B LLM
CCF-A 会议	Zhongkai Yu, Shengwen Liang, Tianyu Ma, Yunke Cai, Ziyuan Nan, Di Huang, Xinkai Song, Yifan Hao, Jie Zhang , Tian Zhi, Yongwei Zhao, Zidong Du, Xing Hu Qi Guo, Tianshi Chen

ToS	Extremely-Compressed SSDs with I/O Behavior Prediction
CCF-A 期刊	Xiangyu Yao, Qiao Li, Kaihuan Lin, Xinbiao Gan, Jie Zhang , Congming Gao, Zhirong Shen, Quanqing Xu, Chuanhui Yang, Jason Xue ACM Transactions on Storage
TC	Land of Oz: Resolving Orderless Writes in Zoned Namespace SSDs
CCF-A 期刊	Yingjia Wang, You Zhou, Fei Wu, Jie Zhang , Ming-Chang Yang IEEE Transactions on Computers
USENIX ATC	ScalaAFA: Constructing User-Space All-Flash Array Engine with Holistic Designs
CCF-A 会议	Shushu Yi, Xiurui Pan, Qiao Li, Qiang Li, Chenxi Wang, Bo Mao, Myoungsoo Jung, Jie Zhang* USENIX Annual Technical Conference
USENIX ATC	ScalaCache: Scalable User-Space Page Cache Management with Software-Hardware
CCF-A 会议	Coordination Li Peng, Yuda An, You Zhou, Chenxi Wang, Qiao Li, Chuanning Chen, Jie Zhang* USENIX Annual Technical Conference
Arxiv	From RDMA to RDCA: Toward High-Speed Last Mile of Data Center Networks Using Remote Direct Cache Access Qiang Li, Qiao Xiang, Derui Liu, Yuxin Wang, Haonan Qiu, Xiaoliang Wang, Jie Zhang , Ridi Wen, Haohao Song, Gexiao Tian, Chenyang Huang, Lulu Chen, Shaozong Liu, Yaohui Wu, Zhiwu Wu, Zicheng Luo, Yuchao Shao, Chao Han, Zongjie WU, Jianbo Dong, Zheng Cao, Jinbo WU, Jiwu Shu, Jiesheng Wu Arxiv
TACO	Characterizing and Optimizing LDPC Performance on 3D NAND Flash Memories
CCF-A 期刊	Qiao Li, Yu Chen, Guanyu Wu, Yajuan Du, Min Ye, Xinbiao Gan, Jie Zhang , Zhirong Shen, Jiwu Shu, Chun Jason Xue ACM Transactions on Architecture and Code Optimization
ISCA	Flagger: Cooperative Acceleration for Large-Scale Cross-Silo Federated Learning
CCF-A 会议	Aggregation Xiurui Pan, Yuda An, Shengwen Liang, Bo Mao, Mingzhe Zhang, Qiao Li, Myoungsoo Jung, Jie Zhang* IEEE/ACM International Symposium on Computer Architecture
ASPLOS	Achieving Near-Zero Read Retry for 3D NAND Flash Memory
CCF-A 会议	Min Ye, Qiao Li, Yina Lv, Jie Zhang , Tianyu Ren, Daniel Wen, Tei-Wei Kuo, Chun Jason Xue ACM International Conference on Architectural Support for Programming Languages and Operating Systems
HPCA	StreamPIM: Streaming Matrix Computation in Racetrack Memory
CCF-A 会议	Yuda An, Yunxiao Tang, Shushu Yi, Li Peng, Xiurui Pan, Guangyu Sun, Zhaochu Luo, Qiao Li, Jie Zhang*

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

IEEE International Symposium on High-Performance Computer Architecture

HPCA

BeaconGNN: Large-Scale GNN Acceleration with Asynchronous In-Storage Computing

CCF-A 会议

Yuyue Wang, Xiurui Pan, Yuda An, [Jie Zhang*](#), Glenn Reinman*

IEEE International Symposium on High-Performance Computer Architecture

HPCA

LearnedFTL: A Learning-based Page-level FTL for Reducing Double Reads in Flash-based SSDs

CCF-A 会议

Shengzhe Wang, Zihang Lin, Suzhen Wu, Hong Jiang, [Jie Zhang](#), Bo Mao

IEEE International Symposium on High-Performance Computer Architecture

HPCA

Midas Touch: Invalid-Data Assisted Reliability and Performance Boost for 3D High-Density Flash

CCF-A 会议

QiaoLi, Hongyang Dang, Zheng Wan, Congming Gao, Min Ye, [Jie Zhang](#), Tei-Wei Kuo, Chun Jason Xue

IEEE International Symposium on High-Performance Computer Architecture

2023

NVMW

Optimizations of Linux Software RAID System for Next-Generation Storage

Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, [Jie Zhang*](#)

The 14th Annual Non-volatile Memories Workshop

SAC

BcBench: Exploring Throughput Processor Designs based on Blockchain Benchmarking

Xiurui Pan, Yue Chen, Shushu Yi, [Jie Zhang*](#)

The 38th ACM/SIGAPP Symposium on Applied Computing

CAL

Intelligent SSD Firmware for Zero-Overhead Journaling

SCI 3 区

Hanyeoreum Bae, Donghyun Gouk, Seungjun Lee, Jiseon Kim, Sungjoon Koh, [Jie Zhang*](#), Myoungsoo Jung*

IEEE Computer Architecture Letters (CAL)

2022

HotStorage

ScalaRAID: Optimizing Linux Software RAID System for Next-Generation Storage

Shushu Yi, Yanning Yang, Yunxiao Tang, Zixuan Zhou, Junzhe Li, Yue Chen, Myoungsoo Jung, [Jie Zhang*](#)

14th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 22)

THPC

Survey on Storage-Accelerator Data Movement

Zixuan Zhou, Shushu Yi, [Jie Zhang*](#)

CCF Transaction on High Performance Computing

NVMW

Integrating New Photonic-Based Heterogeneous Memory into Throughput Accelerators

[Jie Zhang](#), Myoungsoo Jung,

The 13th Annual Non-volatile Memories Workshop

NVMW	HAMS: Hardware Automated Memory-over-Storage for Large-scale Memory Expansion <i>Jie Zhang, Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim, Mahmut Taylan Kandemir, Myoungsoo Jung</i> <i>The 13th Annual Non-volatile Memories Workshop</i>
------	--

2021

MICRO CCF-A 会议	Ohm-GPU: Integrating New Optical Network and Heterogeneous Memory into GPU Multi-Processors <i>Jie Zhang, Myoungsoo Jung,</i> <i>The 54th Annual IEEE/ACM International Symposium on Microarchitecture</i>
ISCA CCF-A 会议	Revamping Storage Class Memory with Hardware Automated Memory-Over-Storage Solution <i>Jie Zhang, Miryeong Kwon, Donghyun Gouk, Sungjoon Koh, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung,</i> <i>The IEEE/ACM International Symposium on Computer Architecture</i>
NVMW	Architecting Throughput Processors with New Flash <i>Jie Zhang, Myoungsoo Jung,</i> <i>The 12th Annual Non-volatile Memories Workshop</i>
NVMW	DRAM-less Accelerator for Energy Efficient Data Processing <i>Jie Zhang, Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung</i> <i>The 12th Annual Non-volatile Memories Workshop</i>
NVMW	Manycore-Based Scalable SSD Architecture Towards One and More Million IOPS <i>Jie Zhang, Miryeong Kwon, Michael Swift, Myoungsoo Jung,</i> <i>The 12th Annual Non-volatile Memories Workshop</i>

2020

ISCA CCF-A 会议	ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis <i>Jie Zhang, Myoungsoo Jung,</i> <i>The IEEE/ACM International Symposium on Computer Architecture</i>
FAST CCF-A 会议	Scalable Parallel Flash Firmware for Many-core Architectures <i>Jie Zhang, Miryeong Kwon, Michael Swift, Myoungsoo Jung,</i> <i>The 18th USENIX Conference on File and Storage Technologies</i>
HPCA CCF-A 会议	DRAM-less: Hardware Acceleration of Data Processing with New Memory <i>Jie Zhang, Gyuyoung Park, David Donofrio, John Shalf, Myoungsoo Jung</i> <i>26th IEEE International Symposium on High-Performance Computer Architecture</i>
ISPASS CCF-C 会议	Data Direct I/O Characterization for Future I/O System Exploration <i>Mohammad Alian, Yifan Yuan, Jie Zhang, Ren Wang, Myoungsoo Jung, Nam Sung Kim</i>

CAL

FastDrain: Removing Page Victimization Overheads in NVMe Storage Stack

SCI-3 区

[Jie Zhang](#), Miryeong Kwon, Sanghyun Han, Nam Sung Kim, Mahmut Kandemir and Myoungsoo Jung

IEEE Computer Architecture Letters (CAL)

2019

HPCA

FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads

CCF-A 会议

[Jie Zhang](#), Myoungsoo Jung, Mahmut Kandemir,
25th IEEE International Symposium on High-Performance Computer Architecture

IISWC

Faster than Flash: An In-Depth Study of System Challenges for Emerging Ultra-Low Latency SSDs

Sungjoon Koh, Junkyeok Jang, Changrim Lee, Miryeong Kwon, [Jie Zhang](#), Myoungsoo Jung,

The 2019 IEEE International Symposium on Workload Characterization

DAC

FlashGPU: Placing New Flash Next to GPU Cores

CCF-A 会议

[Jie Zhang](#), Miryeong Kwon, Hyojong Kim, Hyesoon Kim, Myoungsoo Jung,
The 56th Design Automation Conference (DAC)

TPDS

Exploring Fault-Tolerant Erasure Codes for Scalable All-Flash Array Clusters

CCF-A 期刊

Sungjoon Koh, [Jie Zhang](#), Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,
IEEE Transactions on Parallel and Distributed Systems (TPDS)

NVMW

Addressing Fast-Detrapping for Reliable 3D NAND Flash Design

Mustafa Shihab, [Jie Zhang](#), Myoungsoo Jung, Mahmut Kandemir,

10th Annual Non-Volatile Memories Workshop -- Nominated as Memorable Paper Award

KCC

Maximizing GPU Cache Utilization with Adjustable Cache Line Management

[Jie Zhang](#), Myoungsoo Jung,

Korean Computer Congress (KCC), 2019 -- Nominated as Excellent Paper Award

2018

OSDI

FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for Ultra-Low Latency SSDs

CCF-A 会议

[Jie Zhang](#), Miryeong Kwon, Donghyun Gouk, Changlim Lee, Mohammad Alian, Myoungjun Chun, Mahmut Kandemir, Nam Sung Kim, Jihong Kim, Myoungsoo Jung,
13th USENIX Symposium on Operating Systems Design and Implementation

MICRO

Amber: Enabling Precise Full-System Simulation with Detailed Modeling of All SSD Resources

CCF-A 会议

Donghyun Gouk, Miryeong Kwon, **Jie Zhang**, Sungjoon Koh, Wonil Choi, Nam Sung Kim, Mahmut Kandemir, Myoungsoo Jung,
The 51st Annual IEEE/ACM International Symposium on Microarchitecture

TACO

ReveNAND: A Fast-Drift Aware Resilient 3D NAND Flash Design

CCF-B 期刊

Mustafa Shihab, **Jie Zhang**, Myoungsoo Jung, Mahmut Kandemir,
ACM Transactions on Architecture and Code Optimization (TACO), 2018

Eurosys

FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems

CCF-A 会议

Jie Zhang, Myoungsoo Jung,
The European Conference on Computer Systems (EuroSys), 2018

IPDPS

CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs

CCF-B 会议

Jie Zhang, Shuwen Gao, Nam Sung Kim, Myoungsoo Jung,
32nd IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2018

2017

CAL

SimpleSSD: Modeling Solid State Drive for Holistic System Simulation

SCI-3 区

Myoungsoo Jung, **Jie Zhang**, Ahmed Abulila, Miryeong Kwon, Narges Shahidi, John Shalf, Nam Sung Kim and Mahmut Kandemir,
IEEE Computer Architecture Letters (CAL), 2017

IISWC

Understanding System Characteristics of Online Erasure Coding on Scalable, Distributed and Large-Scale SSD Array Systems

Sungjoon Koh, **Jie Zhang**, Miryeong Kwon, Jungyeon Yoon, David Donofrio, Nam Sung Kim, Myoungsoo Jung,
IEEE International Symposium on Workload Characterization (IISWC), 2017

IISWC

TraceTracker: Hardware/Software Co-Evaluation for Large-Scale I/O Workload Reconstruction

Miryeong Kwon, **Jie Zhang**, Gyuyoung Park, Wonil Choi, David Donofrio, John Shalf, Mahmut Kandemir, Myoungsoo Jung,
IEEE International Symposium on Workload Characterization (IISWC), 2017

NPC

An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing

CCF-C 会议

Jie Zhang, Myoungsoo Jung,
IFIP International Conference on Network and Parallel Computing (NPC), 2017

NPC/IJPP

Enabling Realistic Logical Device Interface and Driver for NVM Express Enabled Full System Simulations

CCF-C 会议

Donghyun Gouk, **Jie Zhang**, Myoungsoo Jung,

2016

- HPCA** **DUANG: Fast and Lightweight Page Migration in Asymmetric Memory Systems**
CCF-A 会议 Hao Wang, [Jie Zhang](#), Gieseok Park, Sharmila Shridhar, Myoungsoo Jung, Nam Sung Kim,
IEEE Symposium on High Performance Computer Architecture (HPCA), 2016
- ASBD** **A Study for Block-level I/O Trace Reconstruction on All-Flash Arrays**
 Miryeong Kwon, [Jie Zhang](#), Gyuyoung Park, Myoungsoo Jung,
Workshop on Architectures and Systems for Big Data (ASBD@ISCA), 2016
- NVMSA** **An In-Depth Study of Next Generation Interface for Emerging Non-Volatile Memories**
 Wonil Choi, [Jie Zhang](#), Shuwen Gao, Jaesoo Lee, Myoungsoo Jung, Mahmut Kandemir,
IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA), 2016
- INFLOW** **ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform
Cache Architecture**
[Jie Zhang](#), Miryeong Kwon, Chanyoung Park, Myoungsoo Jung, Songkuk Kim,
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads
- INFLOW** **Couture: Tailoring STT-MRAM for Persistent Main Memory**
 Mustafa Shihab, [Jie Zhang](#), Shuwen Gao, Josep Sloan, Myoungsoo Jung,
USENIX Workshop on Interactions of NVM/Flash with Operating Systems and Workloads

2015

- ASBD** **CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near
Data Processing**
[Jie Zhang](#), Damian Szmulewicz, Erick Macias, Myoungsoo Jung,
The 5th Workshop on Architecture and System for Big Data (ASBD), 2015
- PACT** **NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing**
CCF-B 会议 [Jie Zhang](#), David Donofrio, John Shalf, Myoungsoo Jung,
The 24th International Conference on Parallel Architecture and Compilation Techniques
- ICCD** **OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory
Characterization**
CCF-B 会议 [Jie Zhang](#), Gieseok Park, David Donofrio, Mustafa Shihab, John Shalf and Myoungsoo Jung,
The 33rd International Conference on Computer Design (ICCD), 2015
- PACT-SRC** **Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing**
[Jie Zhang](#), David Donofrio, John Shalf and Myoungsoo Jung,
International Conference on parallel Architecture and Compilation Techniques (PACT) –

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

ACM SRC 2nd Runner Award, 2015

FAST-WiP **Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing**
Jie Zhang and Myoungsoo Jung,
USENIX Conference on File and Storage Technologies Working in Progress (FAST WiP), 2015

2014

HotStorage **Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration**
Jie Zhang, Myoungsoo Jung,
6th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage 14), 2014

专利发表情况

- “Memory controlling device and computing device including the same”, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, Jie Zhang, America (US20190171566A1)
- “Flash-based accelerator and computing device including the same”, Myoungsoo Jung, Jie Zhang, America (US10824341B2, US20180321859, US20170285968)
- “基于闪存的加速器和包含其的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN107291424)
- “基于闪存的加速器及包括该加速器的计算设备”, Myoungsoo Jung, Jie Zhang, China (CN109460369)
- “Resistance switching memory-based accelerator”, Myoungsoo Jung, Gyuyoung PARK, Jie Zhang, America (US20180321880A1)
- “PARALLEL PROCESSING UNIT, COMPUTING DEVICE INCLUDING THE SAME, AND THREAD SCHEDULING METHOD”, Jie Zhang, Myoungsoo Jung, America (WO2018021620)
- “MEMORY CONTROL APPARATUS AND COMPUTING DEVICE INCLUDING SAME”, JUNG MYOUNGSOO, GOUK DONGHYUN, KWON MIRYEONG, KOH SUNGJOON, 정명수, JIE ZHANG, 국동현, 권미령, 고성준 장지에, Korea (KR1020180126267)
- “COMPUTING DEVICE, METHOD OF PROCESSING INPUT/OUTPUT REQUEST, AND RECORDING MEDIUM”, Jie Zhang, Myoungsoo Jung, Donghyun Gouk, Miryeong Kwon, Sungjoon Koh, America (pending)
- “FLASH-BASED COPROCESSOR”, Jie Zhang, Myoungsoo Jung, America (pending)
- “FLASH STORAGE DEVICE AND METHOD OF SCHEDULING PAGE VICTIMIZATION”, Jie Zhang, Myoungsoo Jung, America (pending)

研究经历

Research Assistant, Computer Architecture and Memory System Lab **2013 年 9 月 – 2021 年 5 月**

- Cache and memory system optimization in GPGPU and multi-core system.
- Non-volatile memory (including Spin-transfer torque magnetic random-access memory and Phase Change Random Access Memory) characterization and optimization.
- Performance, power and thermal optimizations of Solid State Disk (SSD).

学术活动情况

Journal Paper Review/Sub-review

- IEEE Transactions on Computer
- ACM Transactions on Storage
- ACM Transactions on Architecture and Code Optimization

- ACM Transactions on Computer Systems
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Computer Architecture Letters
- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems

Conference Paper Review/Sub-review

- Eurosys'27
- OSDI'26
- MICRO'25'24'18 '16
- ISCA'26'25'24'23
- HPCA'27'26'25'24'18 '16
- ATC'25'23
- ISPASS'26'25
- USENIX ATC'23
- ChinaSys'24'22
- SAC'22
- HotStorage'20
- DAC'20 '19
- NVMSA'17 '16
- ICCD'19 '18 '17 '15
- IPDPS'18 '16
- DATE'19
- ASPLOS'23'22'19 '18 '17

演讲情况

- Invited talk, "ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis", Intel Computational Storage Lab, 2020
- Presentation, "ZnG: Architecting GPU Multi-Processors with New Flash for Scalable Data Analysis", ISCA, online, 2020
- Presentation, "DRAM-less: Hardware Acceleration of Data Processing with New Memory", HPCA, San Diego, CA, 2020
- Presentation, "Scalable Parallel Flash Firmware for Many-core Architectures", FAST, Santa Clara, CA, 2020
- Presentation, "FUSE: Fusing STT-MRAM into GPUs to Alleviate Off-Chip Memory Access Overheads", HPCA, Washington DC, 2019
- Presentation, "FlashGPU: Placing New Flash Next to GPU Cores", DAC, Las Vegas, NV, 2019
- Presentation, "Maximizing GPU Cache Utilization with Adjustable Cache Line Management", Jeju, South Korea, 2019
- Presentation, "FlashShare: Punching Through Server Storage Stack from Kernel to Firmware for Ultra-Low Latency SSDs", OSDI, Carlsbad, CA, 2018
- Presentation, "FlashAbacus: A Self-governing Flash-based Accelerator for Low-power Systems", Eurosys, Porto, Portugal, 2018
- Presentation, "CIAO: Cache Interference-Aware Throughput-Oriented Architecture and Scheduling for GPUs", IPDPS, Vancouver, Canada, 2018
- Presentation, "An In-depth Performance Analysis of Many-Integrated Core for Communication Efficient Heterogeneous Computing", NPC, Anhui, China, 2017

张杰

tel: +8619910031394 | email: jiezh@pku.edu.cn

-
- Presentation, "ROSS: A Design of Read-Oriented STT-MRAM Storage for Energy-Efficient Non-Uniform Cache Architecture", Inflow, Savannah, GA, 2016
 - Presentation, "Couture: Tailoring STT-MRAM for Persistent Main Memory", Inflow, Savannah, GA, 2016
 - Presentation, "CoDEN: A Hardware/Software CoDesign Emulation Platform for SSD-Accelerated Near Data Processing", ASBD, Portland, OR, 2015
 - Presentation, "NVMMU: Direct Solid State Disk Access for GPU-Accelerated Data Processing", PACT, San Francisco, CA, 2015
 - Presentation, "Integrating 3D Resistive Memory Cache into GPGPU for Energy-Efficient Data Processing", PACT SRC, San Francisco, CA, 2015
 - Presentation, "OpenNVM: An Open-Sourced FPGA-based NVM Controller for Low Level Memory Characterization", ICCD, New York city, NY, 2015
 - Presentation, "Shared Non-Volatile Memory Cache for Energy-Efficient High Throughput GPU Computing", FAST WiP, Santa Clara, CA, 2015
 - Presentation, "Power, Energy, and Thermal Considerations in SSD-Based I/O Acceleration", HotStorage, Philadelphia, PA, 2014

教学经历

- Advanced Computer Architecture (Fall'26, Fall'25, Fall'24, Fall'23)
- Computer Architecture (Fall'26, Fall'25, Fall'24, Fall'23, Fall'22)
- Introduction of Computer System (Fall'26, Fall'25, Fall'24, Fall'23, Fall'22)
- IIT 3002 Operating Systems (Fall'15, Fall'16)
- IIT 6036 Computer Organization and Design (Fall'15, Fall'16)
- IIT 7024 Advanced System Architecture (Spring'16)

获奖情况

- 2026: 北京市自然科学基金青年重点项目（原北京杰青）
- 2025: HPCA 名人堂
- 2025: 奥林帕斯先锋奖
- 2023: 英特尔学术英才计划荣誉学者
- 2023: ACM SIGCSE 新星奖
- 2022: 海外优青
- 2021: Our storage-class memory research is selected as KAIST breakthrough 50 years
- 2020-2021: Korean BK21+ Scholarship
- 2020: HPCA travel grant
- 2019: Annual Non-Volatile Memories Workshop (NVMW) -- Nominated as Memorable Paper Award
- 2019: Korea Computer Congress (KCC) -- Best Presentation Paper Award
- 2018: OSDI travel grant
- 2015: ACM Student Research Competition 2nd Runner Award